

The constants do not split invariantly in the Bloch space

Candidate full negative answer to a question in arXiv:2211.06055

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Status: candidate full proof, likely valid. The constant functions have no algebraic complement in the classical Bloch space that is invariant under the Möbius composition action. Invariance under one explicit hyperbolic automorphism already gives a contradiction.

1 Source question

Let

$$\mathcal{B}(\mathbb{D}) = \left\{ f \in \text{Hol}(\mathbb{D}) : \beta(f) := \sup_{z \in \mathbb{D}} (1 - |z|^2) |f'(z)| < \infty \right\}$$

with its usual seminorm, whose nullspace is the one-dimensional space $\mathbb{C}\mathbf{1}$ of constants. Calzi's survey asks whether one can find a $\tilde{U}_0$ -invariant algebraic complement of $\mathbb{C}\mathbf{1}$ in $\mathcal{B}(\mathbb{D})$, and records that the existence of such spaces was unknown [1, p. 3, footnote 1].

Here U_0 is the composition action (up to the harmless choice of inverse), so that $U_0(\psi)f = f \circ \psi^{-1}$ for $\psi \in \text{Aut}(\mathbb{D})$.

It is worthwhile underlying that, whereas the Bloch space is endowed with the corresponding seminorm, and is therefore *not* Hausdorff (otherwise, $\tilde{U}_0$ would not induce a *bounded* representation in $\mathcal{B}(D)$), the space $B^1(D)$ is actually endowed with a norm (e.g., $f \mapsto |f(0)| + |f'(0)| + \|f''\|_{A_1^1(D)}$), and one may find an equivalent norm which is also $\tilde{U}_0$ -invariant. We would like to emphasize that the problem of dealing with non-Hausdorff spaces raises the issue of finding some natural conditions on a given space X which replace the no longer available continuity of the canonical inclusion $X \subseteq \text{Hol}(D)$. It turns out that, as long as one is concerned with maximal spaces, a reasonable condition to impose is the existence of a *decent* continuous linear functional on X , that is, a non-zero continuous linear functional on X which is continuous with respect to the topology of compact convergence; equivalently, a non-zero continuous linear functional on X is decent if it extends to a continuous linear functional on $\text{Hol}(D)$. We shall also say that X is decent if it admits a decent continuous linear functional. Then, $\mathcal{B}(D)$ may be characterized as the maximal *decent* semi-Banach space (i.e., a complete space whose topology is defined by a seminorm) of holomorphic functions in which $\tilde{U}_0$ induces a bounded representation.

Since the minimal space is always a Banach space (for $\lambda \geq 0$), one may avoid these issues and consider only Banach spaces continuously embedded in $\text{Hol}(D)$ when dealing with the minimal space for $\lambda \geq 0$. Notice, though, that $B^1(D)$ is minimal only in the class of the spaces which contain non-constant functions; the space of constant functions is the actual (but rather uninteresting) ‘minimal’ space. If one still wishes to consider semi-Banach spaces, then the decency assumption does not seem to be sufficient, as one might, for instance, find $\tilde{U}_0$ -invariant algebraic complements of the space of constant functions in $\mathcal{B}(D)$, which would then be invariant Banach spaces *which do not embed continuously into* $\text{Hol}(D)$ but are still decent. □ One

¹We do not know whether such spaces exist or not.

Figure 1: The complement question and its “we do not know” footnote on page 3 of the source.

2 Proof intuition

Conjugating one hyperbolic disk automorphism by the Cayley map turns it into multiplication by 2 on the right half-plane. The normalized logarithm of the Cayley map is a Bloch function satisfying the Abel equation $h \circ \phi - h = 1$. An invariant algebraic complement would induce a projection onto constants that commutes with composition by ϕ ; that projection must annihilate the coboundary on the left but fix the constant on the right, an immediate contradiction.

3 Negative answer

Theorem 1. *There is no vector subspace $Y \subseteq \mathcal{B}(\mathbb{D})$ such that*

$$\mathcal{B}(\mathbb{D}) = \mathbb{C}\mathbf{1} \oplus Y$$

algebraically and $f \circ \psi \in Y$ for every $f \in Y$ and every $\psi \in \text{Aut}(\mathbb{D})$. In fact, no such Y exists even if invariance is required only under the cyclic group generated by

$$\phi(z) = \frac{3z+1}{z+3}.$$

Proof. Consider the Cayley map from the disk to the right half-plane,

$$C(z) = \frac{1+z}{1-z}.$$

Multiplication by 2 is an automorphism of the right half-plane, and

$$\phi = C^{-1} \circ (2C), \quad \phi(z) = \frac{3z+1}{z+3}, \quad C(\phi(z)) = 2C(z).$$

Let Log be the principal logarithm. Since $C(\mathbb{D})$ is the right half-plane, the function

$$h(z) = \frac{\text{Log } C(z)}{\log 2}$$

is holomorphic on $\mathbb{D}$. Positive real scaling does not change the argument, so the principal branch gives the exact Abel equation

$$h(\phi(z)) - h(z) = 1 \quad (z \in \mathbb{D}). \tag{1}$$

The function h belongs to the Bloch space. Indeed,

$$h'(z) = \frac{2}{(1-z^2)\log 2},$$

while

$$|1-z^2|^2 - (1-|z|^2)^2 = 4(\text{Im } z)^2 \geq 0.$$

Consequently

$$(1-|z|^2)|h'(z)| \leq \frac{2}{\log 2} \quad (z \in \mathbb{D}).$$

Suppose now that an invariant algebraic complement Y exists, and let $P : \mathcal{B}(\mathbb{D}) \rightarrow \mathbb{C}\mathbf{1}$ be the algebraic projection along Y . Constants are fixed by composition, and invariance under both ϕ and ϕ^{-1} gives $f \circ \phi \in Y$ exactly when $f \in Y$. Hence

$$P(f \circ \phi) = P(f) \quad (f \in \mathcal{B}(\mathbb{D})).$$

Applying P to (1) yields

$$0 = P(h \circ \phi) - P(h) = P(\mathbf{1}) = \mathbf{1},$$

a contradiction. □

4 Strength and scope

The proof is algebraic. It does not assume that Y is closed, that the projection P is continuous, or that an orbit map is measurable. It also uses only one hyperbolic disk automorphism. More generally, any composition-invariant vector space of holomorphic functions which contains $\mathbf{1}$ and the displayed Abel function h has no complement of the constants invariant under ϕ .

This settles the specific existence question on page 3 of the source. It does not settle the separate question later in the survey asking whether every λ -invariant semi-Banach space is ultradecent.

Bounded searches of the local run indexes and the web on 11 August 2026 used the arXiv identifier, the exact complement language, Bloch constants, Möbius invariance, and Abel equations. They found no later answer to the footnote and no matching nonsplitting proof. Novelty confidence is moderate: the construction is elementary and the Abel function itself is classical.

References

- [1] M. Calzi, *A Survey on Invariant Spaces of Holomorphic Functions on Symmetric Domains*, arXiv:2211.06055v3 (2022), <https://arxiv.org/abs/2211.06055>.